

AI Tool Use for Astronomy, and MCP for Data Archives

NASA COSMIC ORIGINS · AI/ML STIG

Fall lecture series on agentic AI · November 9, 2026

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National Radio Astronomy Observatory

CosmicAI

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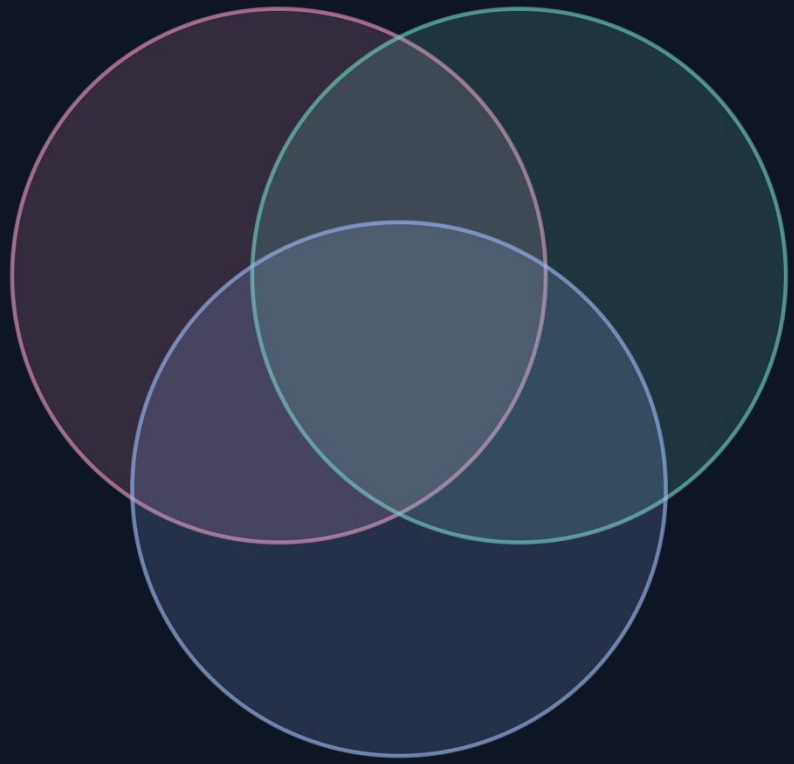
Fall lecture series on agentic AI · Joint session on tool use and MCP

Artificial Intelligence Institute for Cosmic Origins

Funded by the U.S. National Science Foundation and the Simons Foundation

AI Tool Use for Astronomy

The promise of astronomy archives, avoiding the access bottleneck, and lessons learned from the STABLE project



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Slides generated by Claude and modified by Adele; ideas from Adele and the STABLE team. Adele takes full responsibility for the content.

Today’s session: from the why to the how

PART 1 · ADELE PLUNKETT · ≈15 MIN

Tool use for astronomy

- Why archive access is the bottleneck
- Tool use and MCP for astonomy in brief
- What STABLE has built and learned

PART 2 · DAN GAUSE · ≈25 MIN

MCP by example

- Conceiving, building, and testing MANNA
- Demo on how to build an MCP server

Q&A together · 10–15 min

Building on this fall's lectures

Date	Lecture	The thread we pick up today
Sep 14	Speagle · Foundations of LLM agents	An agent is a model with tools; every tool result returns to its context
Sep 21, 28	Stubbs, Saad · Agentic coding and research tools	AI has entered the research space; please use responsibly (https://mathandai.org/). It can accelerate many tasks we do. Use an API when one exists.
Oct 26	Iyer · Retrieval and memory	Ground answers in sources. "Astronomy provides a vital testing ground for these techniques, while also aiding in astrophysical research."
Nov 2	Islam · Skills and benchmarks	Measure agents on real tasks, integrate domain knowledge and technical solutions
Nov 16	Bolliet · Multi-agent orchestration	Next week: agents that coordinate other agents

Today: give agents good tools for the data itself, and test how well they use them.

Multiwavelength science spans many archives

NRAO / ALMA

Radio to submillimeter interferometry

`ALMA Science Archive · TAP / ObsCore`

NOIRLab Astro Data Lab

Optical and IR survey catalogs and images (SMASH, NSC, DES …)

`SQL / ADQL · MyDB · Jupyter`

NASA MAST

UV, optical and near-IR missions (HST, JWST, TESS …)

`TAP · astroquery`

NASA IRSA

Infrared surveys and missions (Spitzer, WISE …)

`TAP · SIA`

NASA HEASARC

X-ray and gamma-ray missions

`TAP · Xamin`

ESO · CADC · Gaia …

Ground- and space-based archives worldwide

`TAP and other IVOA services`

Same sky. Different schemas, column names, units, query services and job systems.

Your favorite here

Catalogs, community-provided, etc.

`?? · ??`

One question, many archive-specific steps

WHAT THE RESEARCHER WANTS

> *Is protostar XYZ inside the VLASS footprint, and does ALMA have any public observations of it? Give the ALMA project codes if there are any. Then, determine the SED of that protostar so I can determine its evolutionary stage.*

Example research question

WHAT IT TAKES TODAY

1. Resolve the protostar's name to coordinates
2. Check the position against the VLASS footprint
3. Query the ALMA archive; list public project codes
4. Find IR to radio photometry across several archives
5. Cross-match; reconcile units, apertures, resolution
6. Build the SED; compute spectral index, Tbol, Lbol
7. Compare with evolutionary classes (0, I, flat, II)

Each step needs archive-specific know-how, and often a different interface.

Where users get stuck

Discovery

Is there data on my source at the frequency, resolution or depth I need, and in which archive?

Query mechanics

Which table, which column, which units? Why does my query return nothing, or far too much?

Retrieval at scale

How do I get the right products for many sources without hitting limits or downloading everything?

Experts move quickly in their home archive. Almost everyone slows down in someone else’s.

THE BOTTLENECK

Archive data are open to everyone, but using them takes fluency in each archive’s dialect.

Tool-using AI agents can carry some of that fluency for us, if we give them the right tools and measure how well they use them.

Tool use turns a model into an agent

Model alone

- Answers from training data, which may be out of date
- Can invent plausible table or column names
- Cannot see what the archive holds today

Model + tools

- Calls functions you define, with typed inputs
- Reads live results (rows, a timeout, zero rows) and adjusts
- Leaves an auditable trail of every query
- **Potentially teaches us methods, sparks new ideas**

Reason → **Act: call a tool** → **Observe the result** *… repeat until done*

Yao et al. 2023 (ReAct, arXiv:2210.03629) · Schick et al. 2023 (Toolformer, arXiv:2302.04761) · Speagle, STIG Sep 14 2026

MCP: a common plug for agents and archives



Build a server once; any MCP-capable client can use it.

IVOA standards (TAP, SIA, SCS, RegTAP) are stable and explicit, a natural fit for tool schemas.

Anthropic 2024, Model Context Protocol (open standard) · modelcontextprotocol.io · Saad, STIG Sep 28 2026 (web agents vs. APIs)

Astronomy agents and tools**: a fast-growing field

Project	What it does	Reference
Mephisto	LLM agents that interpret multi-band galaxy observations via SED fitting	Sun et al. 2024 · arXiv:2409.14807
cmbagent	Multi-agent system for cosmological parameter analysis	Laverick et al. 2024 · arXiv:2412.00431
CTAO agents	Operations and Gammapy analysis-code agents for gamma-ray astronomy	Kostunin et al. 2025 · arXiv:2503.00821
AstroSage	Astronomy-specialized LLM for expert question answering (AstroMLab)	de Haan et al. 2025
AstroVisBench	Benchmark for astronomy computing and visualization code	Joseph et al. 2025 · arXiv:2505.20538
AstroDataBench	Benchmark framework for astronomy archive access	github.com/NSF-Simons-CosmicAI-Institute/AstroDataBench
Quasar	Agent chooses from ~200 tools to search ALMA, NRAO, and more.	quasarassistant.com

*** a non-exhaustive list*

Relatively fewer efforts measure or support cross-archive data discovery: the gap STABLE takes on.

Astronomy agents and tools: a fast-growing field

Project	What it does	Source (GitHub unless noted)
MANNA	IVOA access to Astro Data Lab, ALMA, CADC, ESO…	NSF-Simons-CosmicAI-Institute/manna
Verbinal	Mac app, ~200 tools for CADC and CANFAR	szautkin/canfar-macos
AstroFetch	NASA archives (TAP), ADS, transit plans developed by the NASA Exoplanet Science Institute (NExScI) / Caltech-IPAC.	astrofetch.ipac.caltech.edu (chat) astrofetch.ipac.caltech.edu/mcp
Astro MCP	Unified astroquery and DESI data access	SandyYuan/astro_mcp
NASA MCP	NASA public APIs: APOD, NEOs, space weather	AnCode666/nasa-mcp
wsclean	Radio interferometric imaging with WSClean	d3v-null/wsclean-skill
Astronomy Oracle	Verified catalog data, observing planner	gregario/astronomy-oracle
CelestialMCP	Positions, rise and set times, visibility	Rkm1999/CelestialMCP
ASCOM MCP Server	Telescope, camera, focuser control (Alpaca)	stellarpunk/mcp-server-ascom
astrodb-bot	Skills to build and fill AstroDB databases	astrodbtoolkit/astrodb-bot

Nine MCP servers and one skill, from archive search to imaging to telescope control, listed on the Orrery registry (orrery.science/browse, BJ Fulton), November 2026; credit to each repository's authors.

Relatively fewer efforts measure or support cross-archive data discovery: the gap STABLE takes on.

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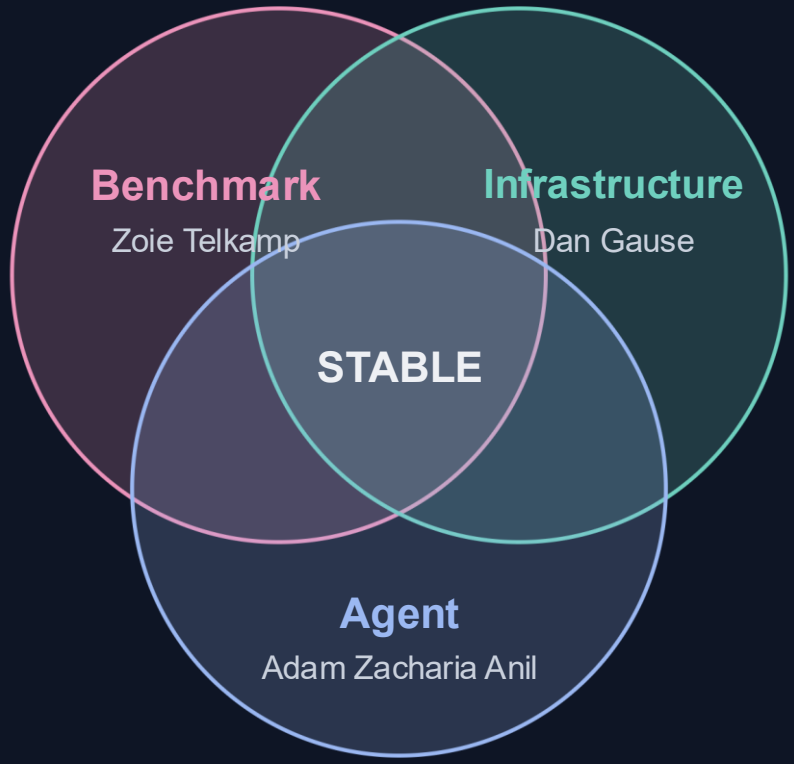
THE PROJECT

STABLE: Summer Team for Astronomical Benchmarking & LLM Engineering

Three summer interns, three complementary projects in parallel, within the NSF–Simons CosmicAI Institute.

Goal: evaluate and build LLM tools for multiwavelength research across major archives and their documentation.

NRAO · NOIRLab · UVA · NYU · UT Austin · with TACC



BENCHMARK · ZOIE TELKAMP

AstroDataBench: measuring data discovery

	Domain expertise suite	Data discovery suite
Tests	Conceptual knowledge of archives and of unfamiliar wavelength regimes	Writing and running code to find data in the ALMA Science Archive and NOIRLab Astro Data Lab
Runs as	The model answers each prompt directly	An agent in a Harbor container with live archive access
Graded by	LLM-as-judge against a gold-standard checklist and rubric	Programmatic tests against a live archive query

Harbor Data discovery runs on Harbor, the open evaluation harness from the Terminal-Bench team: each task is a container with live archive access, and agents and models are swappable and run in parallel. (github.com/harbor-framework/harbor)

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INFRASTRUCTURE · DAN GAUSE

MANNA: an MCP server for astronomy archives

MCP Architecture for NOIRLab and NRAO Archives: IVOA-compliant access to Astro Data Lab, NRAO/ALMA, CADC, ESO, Gaia and more, for any MCP client.

Discover

- `vo_archive_list`
- `vo_registry_search`
- `vo_survey_target`
- `vo_find_observations`

Understand

- `vo_schema_describe`
- `vo_inspect_table`
- `vo_registry_describe`

Query & retrieve

- `vo_target_resolve`
- `vo_tap_query`
- `vo_sia_search`
- `vo_cone_search`

```
pip install manna-mcp
```

Dan takes us inside in Part 2 →

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AGENT · ADAM ZACHARIA ANIL

QUASAR: an AI assistant for radio astronomy

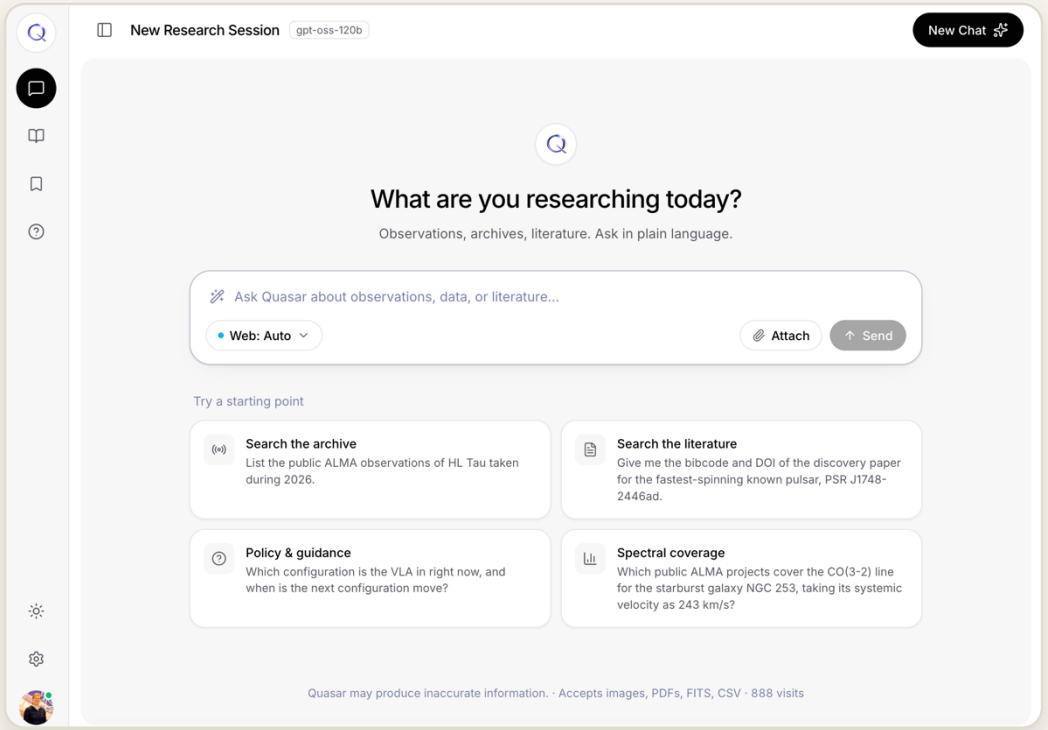
Memory
Remembers a researcher’s interests and past queries

Documentation
Retrieval over technical docs, plus archive tools through MCP

Scale
Recursive Language Models for catalogs larger than the context window

Principles: Integrate existing LLMs and tools rather than reinvent them.
Ask in plain language across observations, archives and literature

quasarassistant.com, deployed at TACC.



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TACC
TEXAS ADVANCED COMPUTING CENTER

What STABLE has shown so far

1 · Curated context beats raw access

Archive quirks (missing columns, enumerated values, index hints) are packaged as knowledge the model reads before it writes a query.

2 · Fewer, smarter tool calls

Composite tools chain resolve → choose archive → search, so the model doesn't have to plan every step.

3 · Keep results bounded

Large results go to async jobs and are fetched client-side, protecting the model's context window.

4 · Grade against the live archive

Executable checks on real queries, with domain knowledge and data discovery scored separately.

A small, coordinated, cross-institution team can work fast and effectively, with tangible deliverables. Keep the project responsive to new technology, tools, and techniques.

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TRY IT YOURSELF

Built during STABLE, open to use

AstroDataBench
Z. Telkamp

Benchmarking framework for multiwavelength data discovery and cross-regime domain knowledge

github.com/NSF-Simons-CosmicAI-Institute/AstroDataBench

MANNA
D. Gause

MCP server for NRAO, NOIRLab and other IVOA archives · pip install manna-mcp

github.com/NSF-Simons-CosmicAI-Institute/manna

QUASAR
A. Zacharia Anil

AI research assistant for radio astronomy

www.quasarassistant.com

Coming soon: Gause et al. (2027, in prep) · Anil et al. (2027, in prep)

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UP NEXT · PART 2

MANNA, hands-on

Dan Gause · NRAO

Inside an astronomy MCP server, and a live tutorial on building one with an agentic coding tool.

Thank you to the STABLE team

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Code: github.com/NSF-Simons-CosmicAI-Institute/AstroDataBench · github.com/NSF-Simons-CosmicAI-Institute/manna · quasarassistant.com

Prior MCP prototypes: github.com/adamzacharia/ALMA_MCP · github.com/SandyYuan/astro_mcp

Harbor Framework Team 2026, Harbor: a framework for evaluating and optimizing agents and models in container environments · github.com/harbor-framework/harbor

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Stubbs, C. 2026 Sep 21, Hands-on I: agentic coding and generative AI research tools

Saad, S. 2026 Sep 28, Hands-on II: agentic coding and research tools (System 1 web agents)

Iyer, K. 2026 Oct 26, Citation grounding, RAG and other ways to make LLMs show their sources

Iyer et al. 2024, Pathfinder · arXiv:2408.01556

Islam, T. 2026 Nov 2, Skills and benchmarks

Bolliet, B. 2026 Nov 16, Multi-agent systems and orchestration

Slides and recordings: ai4astro.org · github.com/nasa-ai-ml-stig/NASA_AI_ML_STIG